

Transformation of state

Tuesday, September 22, 2026

9:42 AM

Orbit in (any) time

Mean anomaly as a function of time: $M - M_0 = n(t - t_0)$

Kepler's equation

$$A(t) = \frac{1}{2}ab(E - e \sin E)$$

$$M = \frac{2A(t)}{ab} = E - e \sin E$$

$M = E - e \sin E$

Relate the eccentric anomaly to true anomaly

$$r^2 = a^2(1 - e^2) \sin^2 E + a^2(e - \cos E)^2,$$

$$\begin{aligned} r^2 &= a^2 \left(1 + e^2 - \underbrace{e^2 \sin^2 E}_{e^2 \cos^2 E} - 2e \cos E \right) \\ &= a^2 (1 - e \cos E)^2 \end{aligned}$$

Take the square root : $r = a(1 - e \cos E)$
Radius-eccentric anomaly equation

$$r = a(1 - e \cos E) = \frac{a(1 - e^2)}{1 + e \cos \theta}$$

$$\cos \theta = \frac{\cos E - e}{1 - e \cos E} \quad (3.10b)$$

Orbit

$$\text{eqn: } r = \frac{p}{1 + e \cos \theta} \rightarrow a(1 - e^2)$$

$$\cos \theta = \frac{\cos E - e}{1 - e \cos E}.$$

This is Curtis eq. (3.10b).

$$\cos E = \frac{e + \cos \theta}{1 + e \cos \theta}$$

This is Curtis eq. (3.10a).

$$\sin \theta = \frac{\sqrt{1 - e^2} \sin E}{1 - e \cos E}.$$

$$\sin E = \frac{\sqrt{1 - e^2} \sin \theta}{1 + e \cos \theta}.$$

This is Curtis eq. (3.11).

$$E = \arctan \left(\sqrt{1 - e^2} \sin \theta, e + \cos \theta \right)$$

$$\theta = \arctan \left(\sqrt{1 - e^2} \sin E, \cos E - e \right)$$

$$\tan^2 \frac{x}{2} = \frac{1 - \cos x}{1 + \cos x}$$

$$\tan^2 \frac{\theta}{2} = \frac{1+e}{1-e} \tan^2 \frac{E}{2}$$

$$\tan \frac{\theta}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{E}{2},$$

$$\tan \frac{E}{2} = \sqrt{\frac{1-e}{1+e}} \tan \frac{\theta}{2}.$$

$$M = E - e \sin E,$$

$$\dot{M} = n = \sqrt{\frac{\mu}{a^3}}$$

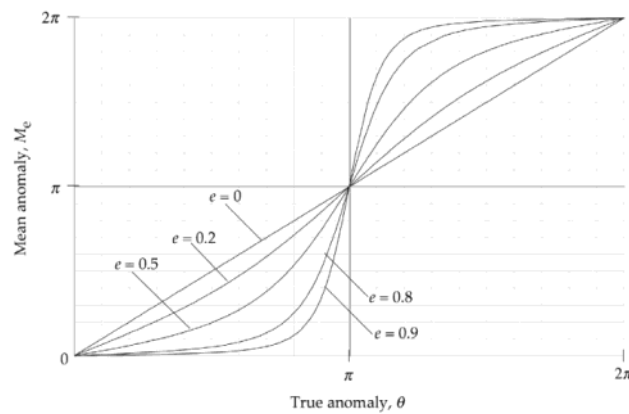
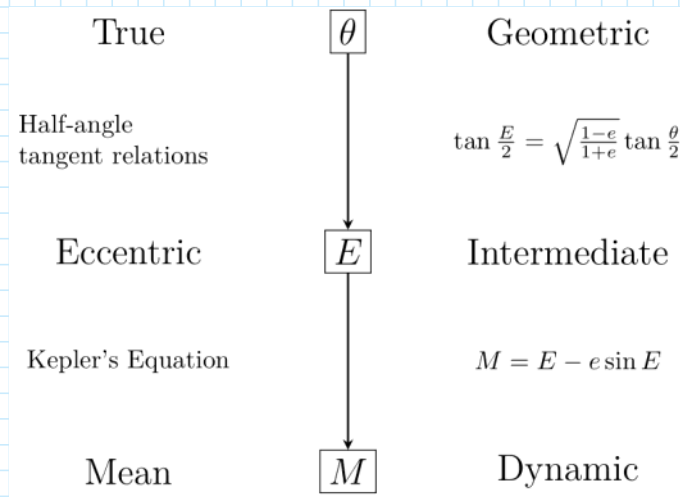


FIG. 3.2
Mean anomaly vs. true anomaly for ellipses of various eccentricities.

Hyperbolic orbit

F (or H) = hyperbolic anomaly $-\infty \leq F \leq +\infty$

$$r = |a| (e \cosh F - 1)$$

$$\tan \frac{\theta}{2} = \sqrt{\frac{e+1}{e-1}} \tanh \frac{F}{2}$$

$$\dot{M} = n = \sqrt{\frac{\mu}{|a|^3}}$$

Parabolic motion

$$r = \frac{1}{2} A^2$$

Parabolic motion

$$D = \tan \frac{\theta}{2}$$

Barker's equation

$$M = D + \frac{1}{3} D^3$$

$$\dot{M} = \frac{1}{(r_p)^{3/2}} \sqrt{\frac{\mu}{2}}$$

radius perigee

Universal variables

Section 3.7

Universal anomaly, Stumpff functions

$$a = \left(\frac{2}{r} - \frac{v^2}{\mu} \right)^{-1}$$

$$\alpha = \frac{1}{a} = \frac{2}{r} - \frac{v^2}{\mu}$$

Universal anomaly

$$\chi = \begin{cases} \sqrt{a} E & \text{ellipse} \\ \frac{h}{\sqrt{\mu}} \tan \frac{\theta}{2} & \text{parabola} \\ \sqrt{|a|} H & \text{hyperbola} \end{cases}$$

$$z = \alpha \chi^2$$

Stumpff functions:

$$S(z) = \sum_{k=0}^{\infty} (-1)^k \frac{z^k}{(2k+3)!} = \frac{1}{6} - \frac{z}{120} + \frac{z^2}{5040} - \frac{z^3}{362,880} + \dots \quad (3.51a)$$

$$C(z) = \sum_{k=0}^{\infty} (-1)^k \frac{z^k}{(2k+2)!} = \frac{1}{2} - \frac{z}{24} + \frac{z^2}{720} - \frac{z^3}{40,320} + \dots \quad (3.51b)$$

$C(z)$ and $S(z)$ are related to the circular and hyperbolic trig functions as follows:

$$S(z) = \begin{cases} \frac{\sqrt{z} - \sin \sqrt{z}}{(\sqrt{z})^3} & (z > 0) \\ \frac{\sinh \sqrt{-z} - \sqrt{-z}}{(\sqrt{z})^3} & (z < 0) \\ \frac{1}{6} & (z = 0) \end{cases} \quad (z = \alpha \chi^2) \quad (3.52)$$

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$$C(z) = \begin{cases} \frac{1 - \cos \sqrt{z}}{z} & (z > 0) \\ \frac{\cosh \sqrt{-z} - 1}{-z} & (z < 0) \\ \frac{1}{2} & (z = 0) \end{cases} \quad (z = \alpha \chi^2) \quad (3.53)$$

Universal Kepler's equation

radial velocity

$$\sqrt{\mu} \Delta t = \frac{r_0 (v_r)_0}{\sqrt{\mu}} \chi^2 C(\alpha \chi^2) + (1 - \alpha r_0) \chi^3 S(\alpha \chi^2) + r_0 \chi \quad (3.49)$$